The Faculty of Medicine of Harvard University
Curriculum Vitae

Date Prepared:	**September 28th, 2025**
Name:	**Xiang Li**
Office Address:	**399 Revolution Drive, 11th Floor, Somerville, MA, 02145**
Work Email:	**xli60@mgh.harvard.edu**

Education:

09/2006	B.E.	Automation	Shanghai Jiaotong University, China
06/2016	PhD	Computer Science, Distinguished Prof. Tianming Liu	University of Georgia

Postdoctoral Training:

08/2016-07/2019	Research Fellow	Medical Image Analysis, Assoc. Prof. Quanzheng Li and Distinguished Prof. James H. Thrall	Harvard Medical School and Massachusetts General Hospital

Faculty Academic Appointments:

08/2019-05/2023	Instructor	Radiology	Harvard Medical School
06/2023-	Assistant Professor	Radiology	Harvard Medical School
09/2023-	Affiliate Faculty Member	Kempner Institute for Natural and Artificial Intelligence	Harvard University

Appointments at Hospitals/Affiliated Institutions:

08/2019-	Research Staff	Radiology	Massachusetts General Hospital

Major Administrative Leadership Positions:

International

2019, 2022-2024	Founding program chair for the *International Workshop on Multiscale Multimodal Medical Imaging*	The Medical Image Computing and Computer-Assisted Intervention Society (MICCAI)
2021	Organizer and program chair for the *International Workshop on Multimodal Learning and Fusion Across Scales for Clinical Decision Support*	The Medical Image Computing and Computer-Assisted Intervention Society (MICCAI)
2024	Organizer and program chair for the *International Workshop on Embodied AI and Robotics for HealTHcare*	The Medical Image Computing and Computer-Assisted Intervention Society (MICCAI)

2024	Organizer and program chair for the *SAGES Critical View of Safety Challenge*	The Medical Image Computing and Computer-Assisted Intervention Society (MICCAI)
2024	Organizer and program chair for the *NeurIPS Workshop on Advancements in Medical Foundation Models*	NeurIPS Foundation
2025	Organizer and program chair for the *AAAI Workshop: AI for Health: Leveraging Artificial Intelligence to Revolutionize Healthcare*	The Association for the Advancement of Artificial Intelligence (AAAI)
2025	Organizer and program chair for the *ISBI Special Session: Unlocking the Potential of Foundation Models in Biomedicine: Opportunities and Challenges*	IEEE Engineering in Medicine and Biology Society (EMBS)

Committee Service:

International

2015	Program Committee	MICCAI Workshop on Machine Learning in Medical Imaging
2017,2023	Program Committee	International Conference on Brain Informatics
2018	Program Committee	Machine Learning in Computational Biology
2019	Program Committee	NeurIPS Workshop on Machine Learning in Computational Biology
2019-2022	Program Committee	ACM SIGKDD Workshop on Mining and Learning from Time Series
2023	Program Committee	International Workshop on Medical Image Learning with Noisy and Limited Data
2023-2024	Area Chair	International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)
2025	Program Committee	IEEE Symposium on Computer-Based Medical Systems
2025	Area Chair	Conference on Neural Information Processing Systems (NeurIPS)
2025	Area Chair	Association for the Advancement of Artificial Intelligence (AAAI) Conference

Grant Review Activities:

2024	ZRG1 CTH-E (57): PAR Panel: Academic-Industrial Partnerships for Translation of Technologies 10/16/2024	National Institutes of Health, Division of Translational and Clinical Sciences (DTCS) Ad hoc reviewer

2025	ZMH1 ERB-E (03) S: BRAIN Initiative: Research on the Ethical Implications of Advancements in Neurotechnology and Brain Science 05/02/2024	National Institutes of Health, BRAIN Initiative Ad hoc reviewer

Editorial Activities:

2021-	Associate Editor	Frontiers in Oncology
2021-	Associate Editor	Frontiers in Radiology
2021-	Associate Editor	Frontiers in Neuroscience
2021	Guest Editor	Frontiers in Neuroscience, Special Issue "Multi-Dimensional Characterization of Neuropsychiatric Disorders"
2022-	Associate Editor	Frontiers in Cardiovascular Medicine
2023-	Associate Editor	Meta-Radiology
2023-	Associate Editor	BMC Biomedical Engineering
2023-	Associate Editor	IEEE Transactions on Artificial Intelligence
2024-	Associate Editor	Data Intelligence
2024	Guest Editor	IEEE Transactions on Neural Networks and Learning Systems, Special Issue "Advancements in Foundation Models"
2024	Associate Editor	Connected Health And Telemedicine
2024	Associate Editor	BMC Artificial Intelligence
2025	Guest Associate Editor	IEEE Transactions on Medical Imaging
2025	Associate Editor	Scientific Report
2025	Ad-hoc Editor	Proceedings of the National Academy of Sciences of the United States of America

Honors and Prizes:

2011	Best Student Paper Award	IEEE International Symposium on Biomedical Imaging	"Brain State Change Detection via Fiber-centered Functional Connectivity Analysis"
2013	Best Student Paper Award	IEEE International Symposium on Biomedical Imaging	"Discovering Common Functional Connectomics Signatures"
2015	Paul D. Coverdell Neuroimaging Franklin Foundation Scholars Program Travel Award	Department of Psychology, University of Georgia	
2015	Cover and Feature Paper	IEEE Transactions on Biomedical Engineering	"Holistic atlases of functional networks and interactions reveal reciprocal organizational architecture of cortical function"

2016	Outstanding Graduate Dissertation/Thesis	University of Georgia	
2018	Most Cited Articles	Journal of the American College of Radiology	“Artificial Intelligence and Machine Learning in Radiology: Opportunities, Challenges, Pitfalls, and Criteria for Success”
2020	Best Paper Awards	IEEE International Symposium on Biomedical Imaging	“ASCNet: Adaptive-Scale Convolutional Neural Networks for Multi-Scale Feature Learning”
2021	MGH Thrall Innovation Grants Award	Massachusetts General Hospital	“Chest Radiographs-based Lung Cancer Screening by the DeepProjection Technique”
2022	Best Paper Awards	IEEE Transactions on Radiation and Plasma Medical Sciences	“Deep Learning-Based Image Segmentation on Multimodal Medical Imaging”
2024	Research Scholar Program	Google Inc.	“Tailoring Large Language Models for the Diagnosis and Management of Late-life Depression Patients with Limited Access to Healthcare Resources”
2024	Distinguished Paper Award	American Medical Informatics Association (AMIA)	“MKRAG: Medical Knowledge Retrieval Augmented Generation for Medical Question Answering”
2025	Featured Article	IEEE Reviews in Biomedical Engineering	“Artificial General Intelligence for Medical Imaging Analysis”
2025	Editor’s Choice Article	Medical Physics	“Fine-tuning open-source large language models to improve their performance on radiation oncology tasks: A feasibility study to investigate their potential clinical applications in radiation oncology”

Report of Regional, National and International Invited Teaching and Presentations

☒ *No presentations below were sponsored by 3rd parties/outside entities*

Regional

2024 “Utilizing Large Language Models for Effective Management and Analysis of Nuclear Medicine Radiology Reports” (Invited Talk)
Hampton Symposium 2024, Boston, MA

2025 “Engineering Large Pre-trained Models: Techniques and Best Practices” (Invited Talk)
School Of Engineering And Applied Sciences, Harvard University

National

2017	"Towards Practical Problems in Deep Learning for Radiology Image Analysis" (Invited Talk) Nvidia GPU Technology Conference, San Jose, CA.
2018	"Deep Learning Algorithm for rapid automatic detection of pneumothorax on chest CT" (selected oral abstract) Annual Meeting of American Roentgen Ray Society, Washington, D.C.
2019	"Personalized Healthcare for Heart Failure with Preserved Ejection Fraction (HFpEF): Diagnosis, Phenotyping and Treatment Optimization with Imaging and EHR Data" (Invited Talk) Department of Statistics, The University of Georgia, Athens, GA.
2022	"Novel Methodologies for Combined Image and EMR Modeling" (Invited Talk) Department of Computer Science and Engineering, The University of Texas at Arlington, Arlington, TX
2022	"Data Governance of the SAGES CVS Challenge" (Invited Talk) Society of American Gastrointestinal and Endoscopic Surgeons, Houston, TX (Virtual)
2023	"Application and Development of Foundational Models in Healthcare" (Invited Talk) Department of Electrical and Computer Engineering, University of Rochester, Rochester, NY (Virtual)
2023	"Application and Development of Foundation Models in Healthcare" (Invited Talk) AIM Seminar Series, Department of Radiation Oncology, UT Southwestern Medical Center, Dallas, TX.
2023	"A New Perspective of Human-Computer Interaction in the Era of Large Pre-trained Models" (Keynote Speech) Workshop on the Intersection of Artificial Intelligence and Human Intelligence (IAIHI), Hoboken, NJ.
2023	"Application and Development of Foundation Models in Healthcare" (Invited Talk) Neuro Image Research and Analysis Laboratories (NIRAL), Department of Psychiatry, University of North Carolina at Chapel Hill, Chapel Hill, NC.
2024	"Development of Specialized Large Language Models for Radiology Report Processing" (Invited Talk) 31st Annual Council on Ionizing Radiation Measurements & Standards Meeting, Rockville, MD.
2024	"Understanding Neurodegenerative Diseases via Multi-modal Data Analytics, Generative Modeling, and Counterfactual Causal Inference" (Invited Talk) The New Investigators in Alzheimer's Disease Meeting, Bethesda, MA
2025	"Generative AI in Healthcare: Foundation, Practice, and Perspectives" (Invited Talk) Department of Computer Science, Stony Brook University, New York, NY

International

2019	"Automated Segmentation of Cervical Nuclei in Pap Smear Images using Deformable Multi-path Ensemble Model" (selected oral full-length paper) IEEE International Symposium on Biomedical Imaging, Venice, Italy.
2023	"Application and Development of Foundational Models in Healthcare" IEEE EMBS Webinar Series, "Frontiers of Biomedical Imaging and Analysis" (Virtual)
2023	"Identification of Causal Relationship between Amyloid-beta Accumulation and Alzheimer's Disease Progression via Counterfactual Inference" Optica Imaging Congress, Boston, MA (selected oral abstract)

2023	“Recent Advances in Sparse and Ultra-Sparse Reconstruction and Generative Modeling for Medical Imaging” Optica Imaging Congress, Boston, MA (invited talk)
2024	“Cross-Modal Retrieval for Alzheimer's Disease Diagnosis Using CLIP-Enhanced Dual Deep Hashing” and “Conditional Score-Based Diffusion Model for Cortical Thickness Trajectory Prediction" AMIA 2024 Annual Symposium, San Francisco, CA (selected oral abstract)
2025	Invited panel talk AI for Medicine and Healthcare, The Association for the Advancement of Artificial Intelligence (AAAI), Philadelphia, PA

Report of Technological and Other Scientific Innovations

AUTOMATED DETECTION AND MANAGEMENT OF VAVLULAR HEART DISEASE USING MACHINE LEARNING	Developing an Aortic Stenosis Ensemble Risk Prediction (AS-ERP) Model. AS-ERP performs Aortic Stenosis patient outcome (length of stay and readmission) prediction based on the input Electronic Medical Records (EMR) data. The model utilizes an ensemble learning scheme consisting of three machine learning classifiers for patient outcome prediction. Internal validation performance meets the clinical acceptance criteria and is superior to the current risk score system developed by the Society of Thoracic Surgeons. US Patent No. 2024/0379239

Report of Scholarship

Peer-Reviewed Scholarship in print or other media:

Research Investigations *indicates co-first authorship, **indicates my mentee

1 J Sun, X Hu, X Huang, Y Liu, K Li, **X Li**, J Han, L Guo, T Liu, J Zhang. Inferring consistent functional interaction patterns from natural stimulus FMRI data. *NeuroImage*. 2012;61(4):987-99.

2 **X Li**, C Lim, K Li, L Guo, T Liu. Detecting brain state changes via fiber-centered functional connectivity analysis. *Neuroinformatics*. 2013;11(2):193.

3 X Zhang, L Guo, **X Li**, T Zhang, D Zhu, K Li, H Chen, J Lv, C Jin, Q Zhao, L Li, T Liu. Characterization of task-free and task-performance brain states via functional connectome patterns. *Medical Image Analysis*. 2013;17(8):1106.

4 **X Li**, D Zhu, X Jiang, C Jin, X Zhang, L Guo, J Zhang, X Hu, L Li, T Liu. Dynamic functional connectomics signatures for characterization and differentiation of PTSD patients. *Human Brain Mapping*. 2014;35(4):1761.

5 J Ou, Z Lian, L Xie, **X Li**, P Wang, Y Hao, D Zhu, R Jiang, Y Wang, Y Chen, J Zhang, T Liu. Atomic dynamic functional interaction patterns for characterization of ADHD. *Human Brain Mapping*. 2014;35(10):5262.

6 D Sabatinelli, D Frank, T Wanger, M Dhamala, B Adhikari, **X Li**. The timing and directional connectivity of human frontoparietal and ventral visual attention networks in emotional scene perception. *Neuroscience*. 2014;277:229.

7 J Zhang*, **X Li***, C Li, Z Lian, X Huang, G Zhong, D Zhu, K Li, C Jin, X Hu, J Han, L Guo, X Hu, L Li, T Liu. Inferring functional interaction and transition patterns via dynamic Bayesian variable partition models. *Human Brain Mapping*. 2014;35(7):3314.

8 X Zhang*, **X Li***, C Jin, H Chen, K Li, D Zhu, X Jiang, T Zhang, J Lv, X Hu, J Han, Q Zhao, L Guo, T Liu. Identifying and characterizing resting state networks in temporally dynamic functional connectomes. *Brain Topography*. 2014;27(6):747.

9 X Jiang, **X Li**, J Lv, T Zhang, S Zhang, L Guo, T Liu. Sparse representation of HCP grayordinate data reveals novel functional architecture of cerebral cortex. *Human Brain Mapping*. 2015;36(12):5301.

10 J Lv*, X Jiang*, **X Li***, D Zhu, H Chen, T Zhang, S Zhang, X Hu, J Han, H Huang, J Zhang, L Guo, T Liu. Sparse representation of whole-brain fMRI signals for identification of functional networks. *Medical Image Analysis*. 2015;20(1):112.

11 J Lv*, X Jiang*, **X Li***, D Zhu, S Zhang, S Zhao, H Chen, T Zhang, X Hu, J Han, J Ye, L Guo, T Liu. Holistic atlases of functional networks and interactions reveal reciprocal organizational architecture of cortical function. *IEEE Transactions on Biomedical Engineering*. 2015;62(4):1120.

12 J Lv, X Jiang, **X Li**, D Zhu, S Zhao, T Zhang, X Hu, J Han, L Guo, Z Li, C Coles, X Hu, T Liu. Assessing effects of prenatal alcohol exposure using group-wise sparse representation of fMRI data. *Psychiatry Research: Neuroimaging*. 2015;233(2):254.

13 M Makkie, S Zhao, X Jiang, J Lv, Y Zhao, B Ge, **X Li**, J Han, T Liu. HAFNI-enabled large-scale platform for neuroimaging informatics (HELPNI). *Brain Informatics*. 2015;2(4):225.

14 J Ou, L Xie, C Jin, **X Li**, D Zhu, R Jiang, Y Chen, J Zhang, L Li, T Liu. Characterizing and differentiating brain state dynamics via hidden Markov models. *Brain Topography*. 2015;28(5):666.

15 J Ou, L Xie, **X Li**, D Zhu, D Terry, A Puente, R Jiang, Y Chen, L Wang, D Shen, J Zhang, L Miller, T Liu. Atomic connectomics signatures for characterization and differentiation of mild cognitive impairment. *Brain Imaging and Behavior*. 2015;9(4):663.

16 Y Hou, T Xiao, S Zhang, X Jiang, **X Li**, X Hu, J Han, L Guo, L Miller, R Neupert, T Liu. Predicting Movie Trailer Viewer's "Like/Dislike" via Learned Shot Editing Patterns. *IEEE Transactions on Affective Computing*. 2016;7(1):29.

17 S Zhang*, **X Li***, J Lv, X Jiang, L Guo, T Liu. Characterizing and differentiating task-based and resting state fMRI signals via two-stage sparse representations. *Brain Imaging and Behavior*. 2016;10(1):21.

18 X Jiang, **X Li**, J Lv, S Zhao, S Zhang, W Zhang, T Zhang, J Han, L Guo, T Liu. Temporal dynamics assessment of spatial overlap pattern of functional brain networks reveals novel functional architecture of cerebral cortex. *IEEE Transactions on Biomedical Engineering*. 2016;65(6):1183.

19 B Ge, M Makkie, J Wang, S Zhao, X Jiang, **X Li**, J Lv, S Zhang, W Zhang, J Han, L Guo, T Liu. Signal sampling for efficient sparse representation of resting state FMRI data. *Brain Imaging and Behavior*. 2016;10:1206.

20 Y Li, H Chen, X Jiang, **X Li**, J Lv, M Li, H Peng, J Tsien, T Liu. Transcriptome Architecture of Adult Mouse Brain Revealed by Sparse Coding of Genome-Wide In Situ Hybridization Images. *Neuroinformatics*. 2017;15(3):285.

21 Y Li, H Chen, X Jiang, **X Li**, J Lv, H Peng, J Tsien, T Liu. Discover mouse gene coexpression landscapes using dictionary learning and sparse coding. *Brain Structure and Function*. 2017;222(9):4253.

22 J Yuan, **X Li**, J Zhang, L Luo, Q Dong, J Lv, Y Zhao, X Jiang, S Zhang, W Zhang, T Liu. Spatio-temporal modeling of connectome-scale brain network interactions via time-evolving graphs. *NeuroImage*. 2017;180:350.

23 B Ge, **X Li**, X Jiang, Y Sun, T Liu. A Dictionary Learning Approach for Signal Sampling in Task-based fMRI for Reduction of Big Data. *Frontiers in Neuroinformatics*. 2018;12.

24 M Makkie*, **X Li***, S Quinn, B Lin, J Ye, G Mon, T Liu. A Distributed Computing Platform for fMRI Big Data Analytics. *IEEE Transactions on Big Data*. 2018;5(2):109.

25 J Thrall, **X Li**, Q Li, C Cruz, S Do, K Dreyer, J Brink. Artificial Intelligence and Machine Learning in Radiology: Opportunities, Challenges, Pitfalls, and Criteria for Success. *Journal of the American College of Radiology*. 2018;15(3):504.

26 W Zhang, J Lv, **X Li**, D Zhu, X Jiang, S Zhang, Y Zhao, L Guo, J Ye, D Hu, T Liu. Experimental Comparisons of Sparse Dictionary Learning and Independent Component Analysis for Brain Network Inference from fMRI Data. *IEEE Transactions on Biomedical Engineering*. 2018;66(1):289.

27 Z Guo**, **X Li***, H Huang, N Guo, Q Li. Deep Learning-based Image Segmentation on Multi-modal Medical Imaging. *IEEE Transactions on Radiation and Plasma Medical Sciences*. 2019;3(2):162.
28 **X Li**, N Guo, Q Li. Functional Neuroimaging in the New Era of Big Data. *Genomics Proteomics and Bioinformatics*. 2019;17(4):393.
29 H Wang, K Xie, L Xie, **X Li**, M Li, C Lyu, H Chen, Y Chen, X Liu, J Tsien, T Liu. Functional Brain Connectivity Revealed by Sparse Coding of Large-Scale Local Field Potential Dynamics. *Brain Topography*. 2019;32(2):255.
30 Y Zhao*, **X Li***, H Huang, W Zhang, S Zhao, M Makkie, M Zhang, Q Li, T Liu. 4D Modeling of fMRI Data via Spatio-Temporal Convolutional Neural Networks (ST-CNN). *IEEE Transactions on Cognitive and Developmental Systems*. 2019;12(3):451.
31 **X Li**, JH Thrall, SR Digumarthy, MK Kalra, PV Pandharipande, B Zhang, C Nitiwarangkul, R Singh, RD Khera, Q Li. Deep learning-enabled system for rapid pneumothorax screening on chest CT. *European Journal of Radiology*. 2019;120:108692.
32 S Jeong*, **X Li***, J Yang, Q Li, V Tarokh. Sparse Representation-Based Denoising for High-Resolution Brain Activation and Functional Connectivity Modeling: A Task fMRI Study. *IEEE Access*. 2020;8:36728.
33 M Zhang**, **X Li***, M Xu, Q Li. Automated Semantic Segmentation of Red Blood Cells for Sickle Cell Disease. *IEEE Journal of Biomedical and Health Informatics*. 2020;24:3095.
34 P Wang, X Jiang, H Chen, S Zhang, **X Li**, Q Cao, L Sun, L Liu, B Yang, Y Wang. Assessing Fine-Granularity Structural and Functional Connectivity in Children with Attention Deficit Hyperactivity Disorder. *Frontiers in Human Neuroscience*. 2020;14:481.
35 X Wang, L Zhang, X Yang, L Tang, J Zhao, G Chen, **X Li**, S Yan, S Li, Y Yang, Y Kang, Q Li, N Wu. Deep Learning Combined with Radiomics May Optimize the Prediction in Differentiating High-Grade Lung Adenocarcinomas in Ground Glass Opacity Lesions on CT Scans. *European Journal of Radiology*. 2020;129:109150.
36 A Zhong*, **X Li***, D Wu, H Ren**, K Kim, Y Kim, V Buch, N Neumark, B Bizzo, WY Tak, SY Park, YR Lee, MK Kang, JG Park, BS Kim, WJ Chung, N Guo, I Dayan, MK Kalra, Q Li. Deep metric learning-based image retrieval system for chest radiograph and its clinical applications in COVID-19. *Medical Image Analysis*. 2021;70:101993.
37 W Xue, J Li, Z Hu, E Kerfoot, J Clough, I Oksuz, H Xu, V Grau, F Guo, M Ng, **X Li**, Q Li, L Liu, J Ma, E Grinias, G Tziritas, W Yan, AMA Labrador, M Garreau, Y Jang, A Debus, E Ferrante, G Yang, T Hua, S Li. Left Ventricle Quantification Challenge: A Comprehensive Comparison and Evaluation of Segmentation and Regression for Mid-ventricular Short-axis Cardiac MR Data. *IEEE Journal of Biomedical and Health Informatics*. 2021;25(9):3541.
38 I Dayan, HR Roth, A Zhong, A Harouni, A Gentili, AZ Abidin, A Liu, AB Costa, BJ Wood, CS Tsai, CH Wang, CN Hsu, CK Lee, P Ruan, D Xu, D Wu, E Huang, FC Kitamura, G Lacey, GC de Antônio Corradi, G Nino, HH Shin, H Obinata, H Ren**, JC Crane, J Tetreault, J Guan, JW Garrett, JD Kaggie, JG Park, K Dreyer, K Juluru, K Kersten, MAB Rockenbach, MG Linguraru, MA Haider, M AbdelMaseeh, N Rieke, PF Damasceno, PMC e Silva, P Wang, S Xu, S Kawano, S Sriswasdi, SY Park, TM Grist, V Buch, W Jantarabenjakul, W Wang, WY Tak, **X Li**, X Lin, YJ Kwon, A Quraini, A Feng, AN Priest, B Turkbey, B Glicksberg, B Bizzo, BS Kim, C Tor-Díez, CC Lee, CJ Hsu, C Lin, CL Lai, CP Hess, C Compas, D Bhatia, EK Oermann, E Leibovitz, H Sasaki, H Mori, I Yang, JH Sohn, KN Murthy, LC Fu, MRF de Mendonça, M Fralick, MK Kang, M Adil, N Gangai, P Vateekul, P Elnajjar, S Hickman, S Majumdar, SL McLeod, S Reed, S Gräf, S Harmon, T Kodama, T Puthanakit, T Mazzulli, VL de Lavor, Y Rakvongthai, YR Lee, Y Wen, FJ Gilbert, MG Flores, Q Li. Federated learning for predicting clinical outcomes in patients with COVID-19. *Nature Medicine*. 2021;27:1735.
39 P You**, **X Li***, Z Wang, H Wang, B Dong, Q Li. Characterization of Brain Iron Deposition Pattern and Its Association with Genetic Risk Factor in Alzheimer's Disease Using Susceptibility-Weighted Imaging. *Frontiers in Human Neuroscience*. 2021;15.

40 P You**, **X Li***, F Zhang, Q Li. Connectivity-based Cortical Parcellation via Contrastive Learning on Spatial-Graph Convolution. *BME Frontiers*. 2022;Article ID 9814824.
41 L Manubens-Gil, Z Zhou, H Chen, A Ramanathan, X Liu, Y Liu, A Bria, T Gillette, Z Ruan, J Yang, M Radojević, T Zhao, L Cheng, L Qu, S Liu, K E Bouchard, L Gu, W Cai, S Ji, B Roysam, C-W Wang, H Yu, A Sironi, D M Iascone, J Zhou, E Bas, E Conde-Sousa, P Aguiar, **X Li**, Y Li, S Nanda, Y Wang, L Muresan, P Fua, B Ye, H-y He, J F Staiger, M Peter, D N Cox, M Simonneau, M Oberlaender, G Jefferis, K Ito, P Gonzalez-Bellido, J Kim, E Rubel, H T Cline, H Zeng, A Nern, A-S Chiang, J Yao, J Roskams, R Livesey, J Stevens, T Liu, C Dang, Y Guo, N Zhong, G Tourassi, S Hill, M Hawrylycz, C Koch, E Meijering, G A Ascoli, H Peng. BigNeuron: a resource to benchmark and predict performance of algorithms for automated tracing of neurons in light microscopy datasets. Nature Methods. 2023;20:824.
42 Z Wang, M He, Y Lv, E Ge, S Zhang, N Qiang, T Liu, F Zhang, **X Li**, B Ge. Accurate Corresponding Fiber Tract Segmentation via FiberGeoMap Learner with Application to Autism. *Cerebral Cortex*. 2023;33(13):8405.
43 J Holmes, Z Liu**, L Zhang, Y Ding, T Sio, L McGee, J Ashman, **X Li**, T Liu, J Shen, W Liu. Evaluating large language models on a highly-specialized topic, radiation oncology physics. *Frontiers in Oncology*. 2023;13.
44 L Zhao, L Zhang, Z Wu, Y Chen, H Dai, X Yu, Z Liu**, T Zhang, X Hu, X Jiang, **X Li**, D Zhu, D Shen, T Liu. When brain-inspired AI meets AGI. *Meta-Radiology*. 2023;1(1).
45 Y Liu, T Han, S Ma, J Zhang, Y Yang, J Tian, H He, A Li, M He, Z Liu**, Z Wu, L Zhao, D Zhu, **X Li**, N Qiang, D Shen, T Liu, B Ge. Summary of ChatGPT-Related Research and Perspective Towards the Future of Large Language Models. **Meta-Radiology**. 2023;1(2).
46 W Liao, Z Liu**, H Dai, S Xu, Z Wu, Y Zhang, X Huang, D Zhu, H Cai, Q Li, T Liu, **X Li**. Differentiate ChatGPT-generated and Human-written Medical Texts. *JMIR Medical Education*. 2023; 9:e48904.
47 J Wang, Z Liu, L Zhao, Z Wu, C Ma, S Yu, H Dai, Q Yang, Y Liu, S Zhang, E Shi, Y Pan, T Zhang, D Zhu, **X Li**, J Jiang, B Ge, Y Yuan, D Shen, T Liu, S Zhang. *Review of Large Vision Models and Visual Prompt Engineering*. Meta-Radiology. 2023; 1(3): 100047.
48 C Liu, Z Liu, J Holmes, L Zhang, L Zhang, Y Ding, P Shu, Z Wu, H Dai, Y Li, D Shen, N Liu, Q Li, **X Li**, D Zhu, T Liu, W Liu. Artificial General Intelligence for Radiation Oncology. *Meta-Radiology*. 2023; 1(3):100045.
49 Z Xiao, Y Chen, J Yao, L Zhang, Z Liu, Z Wu, X Yu, Y Pan, L Zhao, C Ma, X Liu, W Liu, **X Li**, Y Yuan, D Shen, D Zhu, D Yao, T Liu, J Jiang. Instruction-ViT: Multi-modal Prompts for Instruction Learning in Vision Transformer. *Information Fusion*. 2024; 104:102204.
50 X Bi, Y Huang, Z Yang, K Chen, Z Xing, L Xu, **X Li**, Z Liu, T Liu. Structure Mapping Generative Adversarial Network for Multi-view Information Mapping Pattern Mining. *IEEE Transactions on Pattern Analysis and Machine Intelligence*. 2024; 46(4):2252.
51 O Shen, J Pratap, **X Li**, N Chen, A Bhashyam. How Does ChatGPT Use Source Information Compared with Google? A Text Network Analysis of Online Health Information. *Clinical Orthopaedics and Related Research*. 2024; 482(4):578.
52 C Ma, Z Wu, J Wang, S Xu, Y Wei, Z Liu**, F Zeng, X Jiang, L Guo, X Cai, S Zhang, T Zhang, D Zhu, D Shen, T Liu, **X Li**. An Iterative Optimizing Framework for Radiology Report Summarization with ChatGPT. *IEEE Transactions on Artificial Intelligence*. 2024; 5(8):4163.
53 S Kim**, H Ren**, J Charton, J Hu, C Gonzalez, J Khambhati, J Cheng, J DeFrancesco, A Waheed, S Marciniak, F Moura, R Cardoso, B Lima, S McKinney, M Picard, **X Li**, Q Li. Assessment of Valve Regurgitation Severity via Contrastive Learning and Multi-view Video Integration. *Physics in Medicine & Biology*. 2024; 69(4):045020.
54 X Bi, Z Yang, Y Huang, K Chen, Z Xing, L Xu, Z Wu, Z Liu, **X Li**, T Liu. CE-GAN: Community Evolutionary Generative Adversarial Network for Alzheimer's Disease Risk Prediction. *IEEE Transactions on Medical Imaging*. 2024; 43(11):3663–3675.

55 Z Chen, H Ren, Q Li, **X Li**. Motion Correction and Super-Resolution for Multi-slice Cardiac Magnetic Resonance Imaging via an End-to-End Deep Learning Approach. *Computerized Medical Imaging and Graphics*. 2024; 115:102389.
56 Z Liu, L Zhang, Z Wu, X Yu, C Cao, H Dai, N Liu, J Liu, W Liu, Q Li, D Shen, **X Li**, D Zhu, T Liu. Surviving ChatGPT in Healthcare. *Frontiers in Radiology*. 2024; 3:1224682.
57 L Zhang, Z Liu, L Zhang, Z Wu, X Yu, J Holmes, H Feng, H Dai, **X Li**, Q Li, W Wong, S Vora, D Zhu, T Liu, W Liu. Generalizable and Promptable Artificial Intelligence Model to Augment Clinical Delineation in Radiation Oncology. *Medical Physics*. 2024; 51:2187–2199.
58 H Dai, Z Liu, W Liao, X Huang, Y Cao, Z Wu, L Zhao, S Xu, F Zeng, W Liu, N Liu, S Li, D Zhu, H Cai, L Sun, Q Li, D Shen, T Liu, **X Li**. AugGPT: Leveraging ChatGPT for Text Data Augmentation. *IEEE Transactions on Big Data*. 2024; In Press.
59 Z Wu, L Zhang, C Cao, Z Liu, L Zhao, Y Li, X Yu, H Dai, C Ma, G Li, W Liu, Q Li, D Shen, **X Li**, D Zhu, T Liu. Exploring the Trade-Offs: Unified Large Language Models vs Local Fine-Tuned Models for Highly-Specific Radiology NLI Task. *IEEE Transactions on Big Data*. 2024; In Press.
60 K Zhang, R Zhou, E Adhikarla, Z Yan, Y Liu, J Yu, Z Liu, X Chen, B Davison, H Ren, J Huang, C Chen, Y Zhou, S Fu, W Liu, T Liu, **X Li***, Y Chen, L He*, J Zou, Q Li, H Liu, L Sun*. A Generalist Vision-Language Foundation Model for Diverse Biomedical Tasks. *Nature Medicine*. 2024; 30:3129–3141.
61 C Chen, J Miao, D Wu, A Zhong, Z Yan, S Kim, J Hu, Z Liu, L Sun, **X Li**, T Liu, P Heng, Q Li. MA-SAM: Modality-agnostic SAM Adaptation for 3d Medical Image Segmentation. *Medical Image Analysis*. 2024; 98:103310.
62 W Liao, Z Liu, H Dai, Z Wu, Y Zhang, X Huang, Y Chen, X Jiang, D Liu, D Zhu, S Li, W Liu, T Liu, Q Li, H Cai*, **X Li***. Mask-guided BERT for Few-Shot Text Classification. *Neurocomputing*. 2024; 610:128576.
63 W Liao, Z Liu, Y Zhang, X Huang, N Liu, T Liu, Q Li, **X Li***, H Cai*. Zero-shot Relation Triplet Extraction as Next-Sentence Prediction. *Knowledge-Based Systems*. 2024; 304:112507.
64 **X Li**, L Zhao, L Zhang, Z Wu, Z Liu, H Jiang, C Cao, S Xu, Y Li, H Dai, Y Yuan, J Liu, G Li, D Zhu, P Yan, Q Li, W Liu, T Liu, D Shen. Artificial General Intelligence for Medical Imaging Analysis. *IEEE Reviews in Biomedical Engineering*. 2024; In Press.
65 J Lu, S Lu, Y Zhao, L Yang, W-C Chan, J Lian, C-W Lo, M-K Wong, T Li, R Hui, **X Li**, L Xu, J Liang, D Shum. An electronic health record-linked machine learning tool for diabetes risk assessment in adults with prediabetes. *The Innovation Medicine.* 2025; In Press.
66 T Bradshaw, X Tie, J Warner, J Hu, Q Li, **X Li**. Large language models and large multimodal models in medical imaging: a primer for physicians. *Journal of Nuclear Medicine*. 2025; In Press.
67 H Dai**, Y Li**, Z Liu**, L Zhao, Z Wu**, S Song, S Ye, D Zhu, **X Li**, S Li, X Yao, L Shi, T Peng, Q Li, Z Chen, D Zhang, G Mai, T Liu. AD-AutoGPT: An Autonomous GPT for Alzheimer's Disease Infodemiology. *PLOS Global Public Health*. 2025; In Press.
68 S Kim**, P Jin, C Chen, K Kim, Z Lyu, Hui Ren, S Kim, Z Liu**, A Zhong, T Liu, **X Li**, Q Li. MediViSTA: Medical Video Segmentation via Temporal Fusion SAM Adaptation for Echocardiography. *IEEE Journal of Biomedical and Health Informatics*. 2025; In Press.
69 Y Luo, W Li, C Chen, **X Li**, T Liu, T Niu, Y Yuan. LLM-guided Decoupled Probabilistic Prompt for Continual Learning in Medical Image Diagnosis. *IEEE Transactions on Medical Imaging.* 2025; In Press.
70 Z Chen**, S Kim**, H Ren, S Kim, S Yoon, Q Li, **X Li**. Cine Cardiac Magnetic Resonance Segmentation using Temporal-spatial Adaptation of Prompt-enabled Segment-Anything-Model: A Feasibility Study. *Journal of Cardiovascular Magnetic Resonance*. 2025; In Press.
71 Z Liu, Y Li**, P Shu**, A Zhong, H Jinag, Y Pan, L Yang, C Ju, Z Wu, C Ma, C Chen, S Kim**, H Dai**, L Zhao, L Sun, D Zhu, J Liu, W Liu, D Shen, Q Li, T Liu, **X Li**. Radiology-GPT: A Large Language Model for Radiology. *Meta-Radiology.* 2025; In Press.

72 S Kim**, P Jin, S Song, C Chen, Y Li, H Ren, **X Li**, Ti Liu, Q Li. EchoFM: Foundation Model for Generalizable Echocardiogram Analysis. *IEEE Transactions on Medical Imaging*. 2025; In Press.

73 P Wang, Z Liu, Y Li, J Holmes, P Shu, L Zhang, B Laughlin, DS Toesca, **X Li**, Q Li, S Vora, S Patel, T Sio, T Liu, W Liu. Fine-tuning open-source large language models to improve their performance on radiation oncology tasks: A feasibility study to investigate their potential clinical applications in radiation oncology. *Medical Physics*. 2025; In Press.

74 T Zhong, W Zhao, Y Zhang, Y Pan, P Dong, Z Jiang, H Jiang, Y Zhou, X Kui, Y Shang, L Zhao, Li Y, Y Wei, Z Li, J Zhang, L Yang, H Chen, H Zhao, Y Liu, N Zhu, Y Li, Y Wang, J Yao, J Wang, Y Zeng, L He, C Zheng, Z Zhang, M Li, Z Liu, H Dai, Z Wu, L Zhang, S Zhang, X Cai, X Hu, S Zhao, X Jiang, X Zhang, W Liu, **X Li**, D Zhu, L Guo, D Shen, J Han, T Liu, J Liu, T Zhang. ChatRadio-Valuer: A Chat Large Language Model for Generalizable Radiology Impression Generation on Multi-institution and Multi-system Data. *IEEE Transactions on Biomedical Engineering*. 2025; In Press.

75 Y Sun, D Chahine, Q Wen, T Liu, X Li, Y Yuan, F Calamante, J Lv. Voxel-Level Brain States Prediction Using Swin Transformer. *IEEE Journal of Biomedical and Health Informatics*. 2025; In press.

76 Z He, W Li, T Liu, **X Li**, J H, T Z, Y Yi. GAGM: Geometry-aware graph matching framework for weakly supervised gyral hinge correspondence. *Medical Image Analysis*. 2025; In Press.

Other peer-reviewed scholarship: *indicates co-first or co-corresponding authorship, **indicates my mentee

Full-length proceedings of meetings which have undergone a formal peer-review process

1 **X Li**, K Li, L Guo, C Lim, T Liu. Fiber-centered granger causality analysis. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 2011.

2 C Lim*, **X Li***, K Li, L Guo, T Liu. Brain State Change Detection via Fiber-centered Functional Connectivity Analysis. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2011.

3 X Zhang, L Guo, **X Li**, D Zhu, K Li, Z Sun, C Jin, X Hu, J Han, Q Zhao. Characterization of task-free/task-performance brain states. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 2012.

4 S Zhang, **X Li**, J Lv, X Jiang, D Zhu, H Chen, T Zhang, L Guo, T Liu. Sparse representation of higher-order functional interaction patterns in task-based fMRI data. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 2013.

5 S Zhang, J Lv, **X Li**, X Jiang, L Guo, T Liu. Activated Cliques: Network-based Activation Detection in Task-based fMRI. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2013.

6 J Lv, **X Li**, D Zhu, X Jiang, X Zhang, X Hu, T Zhang, L Guo, T Liu. Sparse representation of group-wise fMRI signals. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 2013.

7 P Lv, L Guo, X Hu, **X Li**, C Jin, J Han, L Li, T Liu. Modeling dynamic functional information flows on large-scale brain networks. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 2013.

8 P Wang, D Zhu, **X Li**, H Chen, X Jiang, L Sun, Q Cao, L An, T Liu, Y Wang. Identifying Functional Connectomics Abnormality in Attention Deficit Hyperactivity Disorder. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2013.

9 **X Li**, D Zhu, X Jiang, C Jin, L Guo, L Li, T Liu. Discovering Common Functional Connectomics Signatures. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2013.

10 D Zhu, **X Li**, X Jiang, H Chen, D Shen, T Liu. Exploring High-order Functional Interactions via Structurally-weighted LASSO Models. *International Conference on Information Processing in Medical Imaging (IPMI)* 2013.

11 Z Lian*, **X Li***, J Xing, J Lv, X Jiang, D Zhu, S Zhang, J Xu, M Potenza, T Liu, J Zhang. Exploring Functional Brain Dynamics via a Bayesian Connectivity Change Point Model. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2014.

12 Z Lian, J Lv, J Xing, **X Li**, X Jiang, D Zhu, J Xu, M Potenza, T Liu, J Zhang. Generalized fMRI Activation Detection via Bayesian Magnitude Change Point Model. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2014.
13 Z Lian*, **X Li***, H Zhang, H Kuang, K Xie, J Xing, D Zhu, J Tsien, T Liu, J Zhang. Detecting Cell Assembly Interaction Patterns via Bayesian-based Change-point Detection and Graph Inference Model. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2014.
14 Z Lian*, **X Li***, T Young, Y Hao, J Xing, J Lv, X Jiang, D Zhu, T Liu, J Zhang. Dynamic Network Partition via Bayesian Connectivity Bi-Partition Change Point Model. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2014.
15 S Zhang, X Hu, J Lv, T Zhang, **X Li**, X Jiang, L Guo, T Liu. Learning fMRI-guided Predictor of Video Shot Changes. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2014.
16 C Lv, X Hu, J Han, G Cheng, **X Li**, L Guo, T Liu. Exploring Consistent Functional Brain Networks During Free Viewing of Videos via Sparse Representation. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2014.
17 D Zhu, **X Li**, T Liu. Sparse Representation of Working Memory Processes based on fMRI Data. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2014.
18 X Jiang, J Lv, D Zhu, T Zhang, **X Li**, X Hu, L Guo, T Liu. Discovering Network-level Functional Interactions from Working Memory fMRI Data. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2014.
19 **X Li**, Z Zhou, P Keller, H Zeng, T Liu, H Peng. Interactive exemplar-based segmentation toolkit for biomedical image analysis. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2015.
20 B Ge, J Wang, J Lv, S Zhang, S Zhao, W Zhang, Q Zhao, **X Li**, X Jiang, J Han, L Guo, T Liu. Signal Sampling for Efficient Sparse Representation of Resting-state fMRI Data. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2015.
21 S Zhang*, **X Li***, J Lv, X Jiang, B Ge, L Guo, T Liu. Characterizing and Differentiating Task-based and Resting-state fMRI Signals via Two-stage Dictionary Learning. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2015.
22 X Jiang, **X Li**, J Lv, S Zhao, S Zhang, W Zhang, T Zhang, T Liu. Modeling Functional Dynamics of Cortical Gyri and Sulci. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 2016.
23 **X Li**, M Makkie, B Lin, M Fazli, I Davidson, J Ye, T Liu, S Quinn. Scalable Fast Rank-1 Dictionary Learning for fMRI Big Data Analysis. *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)* 2016.
24 **X Li**, B Lin, J Lv, J Ye, T Liu. Modeling Functional Network Dynamics via Multi-scale Dictionary Learning and Network Continuums. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2016.
25 **X Li**, Q Dong, X Jiang, J Lv, T Liu. Multiple-demand System Identification and Characterization via Sparse Representations of fMRI Data. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2016.
26 C Lyu, **X Li***, J Lv, X Hu, J Han, L Quo, T Liu. Identifying Group-wise Consistent Sub-networks via Spatial Sparse Representation of Natural Stimulus fMRI Data. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2016.
27 M Makkie, **X Li***, T Liu, S Quinn, B Lin, J Ye. Distributed Rank-1 Dictionary Learning: Towards Fast and Scalable Solutions for fMRI Big Data Analytics. *IEEE International Conference on Big Data* 2016.
28 S Zhang, **X Li***, L Guo, T Liu. Exploring Human Brain Activation via Nested Sparse Coding and Functional Operators. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2017.
29 Y Zhao*, **X Li***, M Makkie, S Quinn, B Lin, J Ye, T Liu. Template-guided Functional Network Identification via Supervised Dictionary Learning. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2017.

30 **X Li**, A Zhong, M Lin, N Guo, M Sun, A Sitek, J Ye, J Thrall, Q Li. Self-paced Convolutional Neural Network for Computer Aided Detection in Medical Imaging Analysis. *International Workshop on Machine Learning in Medical Imaging (MLMI)* 2017.
31 S Jeong*, **X Li***, J Yang, Q Li, V Tarokh. Dictionary Learning and Sparse Coding-based Denoising for High-Resolution Task Functional Connectivity MRI Analysis. *International Workshop on Machine Learning in Medical Imaging (MLMI)* 2017.
32 Jiasha Liu*, **X Li***, H Ren**, Q Li. Multi-estimator Full Left Ventricle Quantification Through Ensemble Learning. *International Workshop on Statistical Atlases and Computational Models of the Heart (STACOM)* 2018.
33 Z Guo**, **X Li***, H Huang, N Guo, Q Li. Medical Image Segmentation based on Multi-modal Convolutional Neural Network: Study on Image Fusion Schemes. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2018.
34 Y Zhao*, **X Li***, W Zhang, S Zhao, M Makkie, M Zhang**, Q Li, T Liu. Modeling 4D fMRI Data via Spatio-Temporal Convolutional Neural Networks (ST-CNN). *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 2018.
35 M Zhang**, **X Li***, M Xu, Q Li. Image Segmentation and Classification for Sickle Cell Disease using Deformable U-Net. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 2018.
36 D Wu, K Gong, K Kim, **X Li**, Q Li. Consensus Neural Network for Medical Image Denoising with Only Noisy Training Samples. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 2019.
37 Z Zhou, N Guo, J Cui, X Meng, Y Hu, H Bao, **X Li**, Q Li. Novel Radiomic Features Based on Graph Theory for PET Image Analysis. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2019.
38 X Zhao**, **X Li***, N Guo, Z Zhou, X Meng, Q Li. Multi-Size Computer-Aided Diagnosis Of Positron Emission Tomography Images Using Graph Convolutional Networks. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2019.
39 H Bao, H Ren**, Z Zhou, **X Li**, N Guo, Q Li. 3D Regional Shape Analysis of Left Ventricle Using MR Images: Abnormal Myocadium Detection and Classification. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2019.
40 J Guo**, W Qiu, **X Li**, X Zhao, N Guo, Q Li. Predicting Alzheimer's Disease by Hierarchical Graph Convolution from Positron Emission Tomography Imaging. *IEEE International Conference on Big Data* 2019.
41 W Qiu, J Guo**, **X Li**, M Xu, M Zhang**, N Guo, Q Li. Multi-label Detection and Classification of Red Blood Cells in Microscopic Images. *IEEE International Conference on Big Data* 2020.
42 Q Dong**, N Qiang, J Lv, **X Li**, L Dong, T Liu, Q Li. A Novel fMRI Representation Learning Framework with GAN. *International Workshop on Machine Learning in Medical Imaging (MLMI)* 2020.
43 M Zhang**, J Zhao*, **X Li***, L Zhang, Q Li. ASCNet: Adaptive-Scale Convolutional Neural Networks for Multi-Scale Feature Learning. *IEEE International Symposium on Biomedical Imaging* (ISBI) 2020.
44 Q Dong**, N Qiang, J Lv, **X Li**, T Liu, Q Li. Spatiotemporal Attention Autoencoder (STAAE) for ADHD Classification. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 2020.
45 Q Dong**, N Qiang, J Lv, **X Li**, T Liu, Q Li. Discovering Functional Brain Networks with 3D Residual Autoencoder (ResAE). *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 2020.
46 J Charton**, H Ren**, J Khambhati, J DeFrancesco, J Cheng, A Waheed, S Marciniak, F Moura, R Cardoso, B Lima, E Steen, E Samset, M Picard, **X Li**, Q Li. View Classification of Color Doppler Echocardiography via Automatic Alignment Between Doppler and B-Mode Imaging. *International Workshop on Advances in Simplifying Medical Ultrasound (ASMUS)* 2022.

47 S Rezayi, H Dai, Z Liu**, Z Wu, A Hebbar, A Burns, L Zhao, D Zhu, Q Li, W Liu, S Li, T Liu, **X Li**. ClinicalRadioBERT: Knowledge-Infused Few-Shot Learning for Clinical Notes Named Entity Recognition. *International Workshop on Machine Learning in Medical Imaging (MLMI)* 2022.

48 H Cai, X Huang, Z Liu**, W Liao, H Dai, Z Wu, D Zhu, H Ren**, Q Li, T Liu, **X Li**. Multimodal Approaches for Alzheimer's Detection Using Patients' Speech and Transcript. *Brain Informatics (BI) Conference* 2023.

49 Z Liu**, A Zhong, Y Li, L Yang, C Ju, Z Wu, C Ma, P Shu, C Chen, S Kim**, H Dai, L Zhao, D Zhu, J Liu, W Liu, D Shen, Q Li, T Liu, **X Li**. Tailoring Large Language Models to Radiology: A preliminary approach to LLM adaptation for a highly specialized domain. *International Workshop on Machine Learning in Medical Imaging (MLMI)* 2023.

50 J Charton**, H Ren**, S Kim**, CM Gonzalez, J Khambhati, J Cheng, J DeFrancesco, A Waheed, S Marciniak, F Moura, R Cardoso, B Lima, M Picard, **X Li**, Q Li. Multi-task Learning for Hierarchically-Structured Images: Study on Echocardiogram View Classification. *International Workshop on Advances in Simplifying Medical Ultrasound (ASMUS)* 2023.

51 H Dai**, M Hu, Q Li, L Zhang, L Zhao, D Zhu, I Diez, J Sepulcre, F Zhang, X Gao, M Liu, Q Li, S Li, T Liu, **X Li**. Graph-Based Counterfactual Causal Inference Modeling for Neuroimaging Analysis. *International Workshop on Multiscale Multimodal Medical Imaging (MMMI)* 2023.

52 B Kim, J Choi, E Yun, K Kim, **X Li**, Juho Lee. Large-scale Graph Representation Learning of Dynamic Brain Connectome with Transformers. *NeurIPS Workshop on Temporal Graph Learning* 2023.

53 Y Shi, H Ma, W Zhong, Q Tan, G Mai, **X Li**, T Liu, J Huang. Chatgraph: Interpretable Text Classification by Converting ChatGPT Knowledge to Graphs. *IEEE International Conference on Data Mining Workshops (ICDMW)* 2023.

54 W Liao, Z Liu, Y Zhang, X Huang, F Qi, S Ding, H Ren, Z Wu, H Dai, S Li, L Wu, N Liu, Q Li, T Liu, **X Li**, H Cai. Coarse-to-fine Knowledge Graph Domain Adaptation based on Distantly-supervised Iterative Training. *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)* 2023.

55 S Yoon**, J Pratap, W C Liu, M Tivnan, H Ren, A Bhashyam, Q Li, N Chen, **X Li**. High-resolution 3D CT synthesis from bidirectional X-ray Images using 3D Diffusion Model. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2024.

56 B Kim, J Choi, E Yun, K Kim, **X Li**, J Lee. Learning Dynamic Brain Connectome with Graph Transformers for Psychiatric Diagnosis Classification. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2024.

57 Z Chen**, H Ren, Q Li, **X Li**. Motion Correction and Super-Resolution for Multi-slice Cardiac Magnetic Resonance Imaging via a Multi-stage Deep Learning Approach. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2024.

58 J Pratap, S Yoon**, W C Liu, Q Li, A Bhashyam, N Chen, **X Li**. Zero-Shot Novel View Synthesis of Wrist X-Rays using Latent Diffusion Model. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2024.

59 W Chen**, L Shen, J Lin, J Luo, **X Li***, Y Yuan*. Fine-Grained Image-Text Alignment in Medical Imaging Enables Cyclic Image-Report Generation. *Annual Meeting of the Association for Computational Linguistics (ACL)* 2024.

60 Y Huang, L Sun, H Wang, S Wu, Q Zhang, Y Li, C Gao, Y Huang, W Lyu, Y Zhang, X Li, H Sun, Z Liu, Y Liu, Y Wang, Z Zhang, B Vidgen, B Kailkhura, C Xiong, C Xiao, C Li, E P Xing, F Huang, H Liu, H Ji, H Wang, H Zhang, H Yao, M Kellis, M Zitnik, M Jiang, M Bansal, J Zou, J Pei, J Liu, J Gao, J Han, J Zhao, J Tang, J Wang, J Vanschoren, J Mitchell, K Shu, K Xu, K-W Chang, L He, L Huang, M Backes, N Z Gong, P S Yu, P-Y Chen, Q Gu, R Xu, R Ying, S Ji, S Jana, T Chen, T Liu, T Zhou, W Y Wang, **X Li**, X Zhang, X Wang, X Xie, X Chen, X Wang, Y Liu, Y Ye, Y Cao, Y Chen, Y Zhao. Position Paper: TrustLLM: Trustworthiness in Large Language Models. *International Conference on Machine Learning (ICML)* 2024.

61 Z He, W Li, Y Jiang, Z Peng, P Wang, **X Li**, T Liu, J Han, T Zhang, Y Yuan. F2TNet: FMRI to T1w MRI Knowledge Transfer Network for Brain Multi-phenotype Prediction. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 2024.

62 S Yoon**, R Hu, Y Wang, M Tivnan, Y D Son, D Wu, **X Li**, K Kim, Q Li. Efficient Volumetric Conditional Score-based Residual Diffusion Model for PET Denoising. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 2024.

63 Q Xiao**, S Yoon**, H Ren, M Tivnan, L Sun, Q Li, T Liu, Y Zhang, **X Li**. Conditional Score-Based Diffusion Model for Cortical Thickness Trajectory Prediction. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 2024.

64 W Chen**, P Wang, H Ren, L Sun, Q Li, Y Yuan, **X Li**. Medical Image Synthesis via Fine-Grained Image-Text Alignment and Anatomy-Pathology Prompt. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 2024.

65 M Tivnan, S Yoon**, Z Chen, **X Li**, D Wu, Q Li. Hallucination Index: An Image Quality Metric for Generative Reconstruction Models. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 2024.

66 Y Shi**, S Xu, T Yang, Z Liu, T Liu, **X Li***, N Liu*. MKRAG: Medical Knowledge Retrieval Augmented Generation for Medical Question Answering. *Annual Symposium of American Medical Informatics Association (AMIA)* 2024.

67 Z Chen, S Kim, H Ren, Q Li, **X Li**. Temporal-spatial Adaptation of Promptable SAM Enhances Accuracy and Generalizability of Cine CMR Segmentation. *International Workshop on Foundation Models for General Medical AI (MedAGI)* 2024.

68 C Ma, H Jiang, W Chen, Y Li, Z Wu, X Yu, Z Liu, L Guo, D Zhu, T Zhang, D Shen, T Liu, **X Li**. Eye-gaze Guided Multi-modal Alignment for Medical Representation Learning. *Annual Conference on Neural Information Processing Systems (NeurIPS)* 2024.

69 H Cui, L Mao, X Liang, J Zhang, H Ren, Q Li, **X Li***, C Yang*. Biomedical Visual Instruction Tuning with Clinician Preference Alignment. *Annual Conference on Neural Information Processing Systems (NeurIPS)* 2024.

70 Y Pan, H Jiang, J Chen, Y Li, H Zhao, Y, P, Z Wu, Z Liu**, D Zhu, **X Li**, Y Abate, T Liu. EG-SpikeFormer: Eye-Gaze Guided Transformer on Spiking Neural Networks for Medical Image Analysis. *IEEE International Symposium on Biomedical Imaging (ISBI)* 2025.

71 Y Shi**, Q Li, J Sun, **X Li***, N Liu*. Enhancing Cognition and Explainability of Multimodal Foundation Models with Self-Synthesized Data. *International Conference on Learning Representations (ICLR)* 2025.

72 Y Li**, S Kim, Z Wu, H Jiang, Y Pan, P Jin, S Song, Y Shi, X Yu, T Yang, T Liu, Q Li, **X Li**. ECHOPulse: ECG Controlled Echocardiogram Video Generation. *International Conference on Learning Representations (ICLR)* 2025.

73 Y Oh, P Jin, S Park, S Kim**, S Yoon**, J Kim, K Kim, **X Li***, Q Li*. Distribution-aware Fairness Learning in Medical Image Segmentation from a Control-Theoretic Perspective. *International Conference on Machine Learning (ICML)* 2025.

74 S Yoon, Y Oh, **X Li**, Y Xin, M Cereda, Q Li. High-fidelity 3D Lung CT Synthesis in ARDS Swine Models using Score-based 3D Residual Diffusion Models. *SPIE Medical Imaging* 2025.

75 W Wang, Z Ma, Z Wang, C Wu, J Ji, W Chen, **X Li**, Y Yuan. A Survey of LLM-based Agents in Medicine. *Annual Meeting of the Association for Computational Linguistics (ACL)* 2025.

76 S Yoon, S Song, P Jin, M Tivnan, Y Oh, S Kim, D Wu, **X Li**, Q Li. Cascaded 3D Diffusion Models for Whole-body 3D 18-FDG PET/CT synthesis from Demographics. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 2025.

77 Z Yan**, S Song, D Song, Y Li, R Zhou, W Sun, Z Chen, S Kim, H Ren, T Liu, Q Li, **X Li**, L He, L Sun. SAMed-2: Segment Anything in Medical Images via Temporal Awareness and Confidence-Driven Memory. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 2025.

78 R Meng, S Song, P Jin, Y Oh, Lin T, Y Wang, Y Sun, L Chen, **X Li**, N Guo, Q Li, D Shen. MAST-Pro: Dynamic Mixture-of-Experts for Adaptive Segmentation of Pan-Tumors with Knowledge-Driven Prompts. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 2025.

Non-peer reviewed scholarship in print or other media:

Reviews, chapters, and editorials

1. Wang P, Zhao S, **Li X**, Lv J. Editorial: Multi-dimensional characterization of neuropsychiatric disorders, *Frontiers in Neuroscience* 2022.
2. J Zhu, M Xu, Y Jin, A Novak, S Ather, BW Papiez, **X Li**, J Wu. AAAI Bridge Program 2025: AI for Medicine and Healthcare, *Proceedings of Machine Learning Research* 2025.

Books/textbooks for the medical or scientific community

1. Wu Z, Shu P, Li Y, Liu T, **Li X**. Robot Control via Natural Instructions Empowered by Large Language Model, *In Book: Emerging Frontiers in Human-Robot Interaction*, Springer Nature 2024.

Other non-peer reviewed scholarship

1. Barron D, **Li X,** Li Q. Editor's Blog: Can AI Solve the Clinical Data Problem? *Science* 2024.